HIMU6

6-axis inertial measurement unit (3-axis accelerometer + 3-axis gyroscope)

Features

- Accelerometer ranges: ±2 g, ±4 g, ±8 g, ±16 g
- Gyroscope ranges: ±125, ±250, ±500, ±1000 and ±2000 dps
- Output data rate up to 6.4 kHz
- Accelerometer noise density: 90 µg/√Hz
- Gyroscope noise density: 0.007 dps/√Hz
- Supply VDD: 1.71 V to 3.6 V; VDDIO: 1.2 V to 3.6 V
- Current in normal mode (both sensors): 0.9 mA
- I²C up to 1 MHz (address 0x6A, or 0x6B when SDO is high) and SPI up to 10 MHz
- Two programmable interrupt pins
- LGA-14 package, 2.5 × 3.0 × 0.83 mm

General description

The HIMU6 combines a MEMS accelerometer and gyroscope with a configurable FIFO, tap/step detection and motion-wake features for low-power motion sensing.

Applications

Wearables, Drones, Gaming, Industrial vibration monitoring.

Ordering information

Order code	Package	Packing	Temperature grade
HIMU6-LGA14-T	LGA-14	Tape & reel, 5000	–40 °C to 85 °C

Absolute maximum ratings

Stresses beyond these ratings may cause permanent damage. Functional operation is not implied.

Parameter	Min	Max	Unit
VDD to GND	–0.3	4.0	V
Acceleration (any axis, 0.2 ms)	—	10000	g
Storage temperature	–40	125	°C

Recommended operating conditions

Parameter	Min	Typ	Max	Unit
Supply voltage VDD	1.71	1.8	3.6	V
I/O supply VDDIO	1.2	1.8	3.6	V
Operating temperature	–40	—	85	°C

Electrical characteristics

Typical values at TA = 25 °C unless otherwise noted.

Parameter	Conditions	Min	Typ	Max	Unit
Accelerometer noise density	±4 g range	—	90	—	µg/√Hz
Gyroscope noise density	±2000 dps range	—	0.007	—	dps/√Hz
Current, both sensors	Normal mode, ODR 104 Hz	—	0.9	1.1	mA
Current, accelerometer only	Low-power mode	—	25	—	µA
Maximum output data rate		—	6.4	—	kHz
I²C address (SDO = GND)	7-bit		0x6A		
I²C address (SDO = VDDIO)	7-bit		0x6B		

Pin configuration and functions

Pin	Name	Type	Description
1	SDO/SA0	Input	SPI data out / I²C address bit.
2	SDx	I/O	Aux sensor data.
4	INT1	Output	Programmable interrupt 1.
5	VDDIO	Power	I/O supply.
8	VDD	Power	Core supply.
13	SCL/SPC	Input	I²C clock / SPI clock.
14	SDA/SDI	I/O	I²C data / SPI data in.

Package and thermal information

Package	Thermal resistance θJA	Dimensions
LGA-14	θJA = 90 °C/W	2.5 × 3.0 mm

Application information

Mount the sensor away from components that generate mechanical stress. Calibrate zero-g offset after reflow.

Revision history

Revision	Date	Change
Rev A	2024-10-05	Initial release
Rev B	2025-03-11	Added 6.4 kHz output data rate